

Database Requirement Analysis:

1 Users of the System

The system is designed for the following users:

- Admin
- ESG Analyst
- Company Representative

2 Data Requirements

The system stores and processes the following data:

- Company Information
- ESG Reports
- ESG Scores
- Risk Prediction Results
- AI Generated Summaries
- User Account Information

3 Functional Requirements

- Upload ESG Reports
- Calculate ESG Scores
- Predict ESG Risk Levels
- Generate AI-Based Report Summaries
- View Company ESG Analytics
- Manage User Accounts

4 Non-Functional Requirements

- Secure Data Storage
- Fast Processing of Reports
- Data Accuracy and Integrity
- Scalability for Multiple Companies

5 Entities Identified

- Company
- ESG_Report
- ESG_Score
- Risk_Prediction
- Report_Summary
- User

6 Attributes

Company:

company_id, company_name, industry, country, website

ESG_Report:

report_id, company_id, report_year, report_file, upload_date

ESG_Score:

score_id, report_id, environmental_score, social_score, governance_score, total_esg_score

Risk_Prediction:

risk_id, report_id, risk_level, risk_score

Report_Summary:

summary_id, report_id, summary_text

User:

user_id, username, email, password, role